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CC Exp

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$f(x,y)$ = no of different bits in their representation
Return the sum of $f(a_i, a_j)$ for all $0<i<j<n$ pairs in assay a.

☆ Approach :-

→ We will check for 0 to 31st bit as numbers are in integer range.
→ For i^{th} bit we will count the number of elements which has their i^{th} bit set and unset, then
Sum += cumBitSum. (setCount * unsetCount)
→ We will do thin for all 32 bits.
→ As order matters, but all pairs are 'just mirrors. If we charge ordes, ∴ return sum*2 % (1e947)

Code:

```
public static int solve (int[] ass) {
    int MOD = (int) 1e9+7;
    int sum=0;
    for (int i=0; i<32; i++) {
        int setCount =0, unsetCount =0;
        for (int j=0; j<aser.length; j++) {
            if ((arr[j] & (1<<i)) != 0) {
                setCount ++;
            } else {
                unSetCount ++;
            }
        }
        Sum = (((setCount % MOD) * (unsetCount % MOD)) % MOD
                                    + Sum) % MOD
    }
```

```
    return sum*2;
}
```

* Test Cases !

input 1 + [1,3,5]

0	0	1
0	1	1
1	0	1

sum = 0

<u>Dry Run</u> 0th bit → setCount = 3, unset = 0

Sum = sum + 3×0 = sum = 0

1st bit → setCount = 1, unsetCount = 2

∴ Sum = 2

2nd bit → setCount = 1, unsetCount = 2

∴ sum = 4

return sum*2 = 8

* Time Complexity

$O(32*n) \Rightarrow O(n)$

Space Complexity.

$O(1)$ As we didn't use any additional space.